JEE Mains 2020 Jan Chapter wise Question Bank

Magnetic Effects of Current

Q1

A current carrying circular loop is placed in an infinite plane if ϕ_i is the magnetic flux through the inner region and ϕ_o is magnitude of magnetic flux through the outer region, then
एक धारावाही वृत्ताकार लूप अनन्त तल में स्थित है, यदि आन्तरिक क्षैत्र से सम्बंधित चुम्बकीय फलक्स ϕ_i तथा बाह्य क्षैत्र से सम्बंधित चुम्बकीय फलक्स ϕ_o हो तो

(1) $\phi_i > \phi_o$
(2) $\phi_i < \phi_o$
(3) $\phi_i = -\phi_o$
(4) $\phi_i = \phi_o$

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3
As magnetic field lines always form a closed loop, hence every magnetic field line creating magnetic flux in the inner region must be passing through the outer region. Since flux in two regions are in opposite direction,
$\therefore \quad \phi_i = -\phi_o$

Q2

Consider a loop ABCDEFA with coordinates A (0, 0, 0), B(5, 0, 0), C(5, 5, 0), D(0, 5, 0) E(0, 5, 5) and F(0, 0, 5). Find magnetic flux through loop due to magnetic field $\vec{B} = 3\hat{i} + 4\hat{k}$

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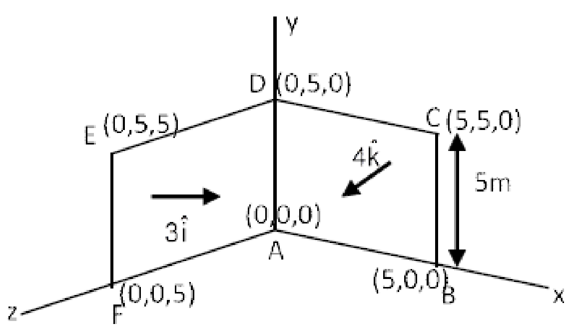
175
$\phi = \vec{B}.\vec{A} = (3\hat{i} + 4\hat{k}).(25\hat{i} + 25\hat{k})$

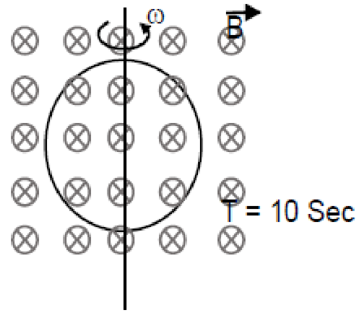


$\phi = (3 \times 25) + (4 \times 25) = 175$ weber

Q3

A ring is rotated about diametric axis in a uniform magnetic field perpendicular to the plane of the ring. If initially the plane of the ring is perpendicular to the magnetic field. Find the instant of time at which EMF will be maximum & minimum respectively :

एक वलय एकसमान चुम्बकीय क्षेत्र (वलय के तल के लम्बवत्) में व्यास के अनुदिश घुर्णन कर रहा है। प्रारम्भिक में वलय का तल चुम्बकीय क्षेत्र के लम्बवत् है। वह समय जब EMF का मान क्रमशः अधिकतम तथा न्यूनतम होगा



at t = 0

(1) 2.5 sec, 5 sec (2) 5 sec, 7.5 sec (3) 2.5 sec, 7.5 sec (4) 10 sec, 5 sec

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(1)

$\therefore \quad \omega = \frac{2\pi}{T} = \frac{\pi}{5}$

When $\omega t = \frac{\pi}{2}$

$\because \quad \phi$ will be minimum.

$\therefore \quad$ e will be maximum

$$t = \frac{\frac{\pi}{2}}{\frac{\pi}{5}} = 2.5\,\text{sec}$$

When $\omega t = \pi$

$\because \quad \phi$ will have maximum.

$\because \quad$ e will be minimum.

$$t = \frac{\pi}{\pi/5} = 5\ \text{sec.}$$

Q4

Electric field in space is given by $\vec{E}(t) = E_0 \frac{(\hat{i}+\hat{j})}{\sqrt{2}}$ cos(ωt + Kz). A positively charged particle at (0, 0, π/K) is given velocity $v_0\hat{k}$ at t = 0. Direction of force acting on particle is

(1) f = 0

(2) Antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(3) Parallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(4) $\hat{k}$

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(2)

Force due to electric field is in direction $-\frac{(\hat{i}+\hat{j})}{\sqrt{2}}$

because at t = 0, E = $-\frac{(\hat{i}+\hat{j})}{\sqrt{2}}$ E_0

Force due to magnetic field is in direction $q(\vec{v} \times \vec{B})$ and $\vec{v} \parallel \hat{k}$

$\therefore$ it is parallel to $\vec{E}$

$\therefore$ net force is antiparallel to $\frac{(\hat{i}+\hat{j})}{\sqrt{2}}$.

Q5

When a proton of KE = 1.0 MeV moving in South to North direction enters the magnetic field (from West to East direction), it accelerates with acceleration a = 10^{12} m/s^2. The magnitude of magnetic field is–

जब गतिज ऊर्जा KE = 1.0 MeV का एक प्रोटोन दक्षिण से उत्तर की तरफ गति करता हुआ एक चुम्बकीय क्षेत्र (पश्चिम से पूर्व की तरफ निर्देशित), में प्रवेश करता है तो यह **a** = 10^{12} m/s^2 त्वरण से त्वरित होता है। चुम्बकीय क्षेत्र का परिमाण ज्ञात करो।

(1) 0.71 mT (2) 7.1 mT (3) 71 mT (4) 710 mT

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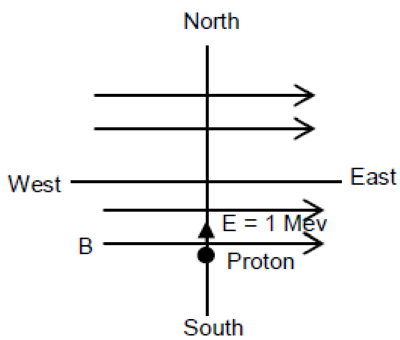
(1)



$\because$ K.E. = $1.6 \times 10^{-13} = \frac{1}{2} \times 1.6 \times 10^{-27} v^2$

$v = \sqrt{2} \times 10^7$ m/s

$\therefore$ Bqv = ma

$B = \frac{1.6 \times 10^{-27} \times 10^{12}}{1.6 \times 10^{-19} \times \sqrt{2} \times 10^7}$

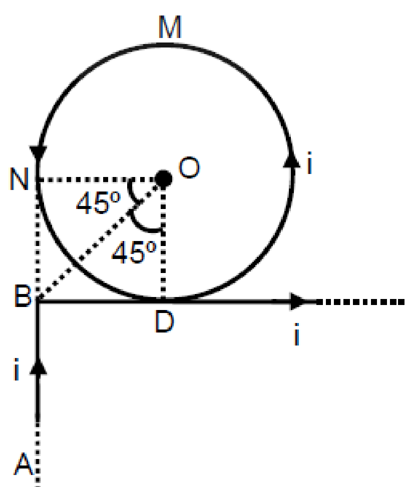
$= 0.71 \times 10^{-3}$ T

So, 0.71 mT

Q6

Find magnetic field at O.

O पर चुम्बकीय क्षेत्र ज्ञात किजिये।



(1) $\frac{\mu_0 i}{2\pi R}\left[\frac{-1}{\sqrt{2}} + \pi\right]$ (2) $\frac{\mu_0 i}{2\pi R}[\pi - 1]$ (3) $\frac{\mu_0 i}{2R}$ (4) $\frac{\mu_0 i}{2\pi R}\left[\frac{1}{\sqrt{2}} + \pi\right]$

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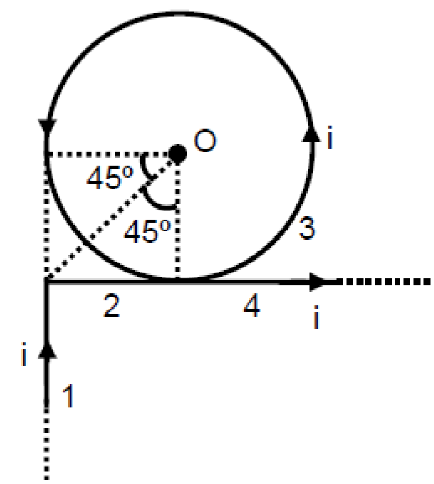
(4)



$\vec{B}_0 = (\vec{B}_0)_1 + (\vec{B}_0)_2 + (\vec{B}_0)_3 + (\vec{B}_0)_4$

$\frac{\mu_0 i}{4\pi R}[\sin 90^\circ - \sin 45^\circ] \otimes + \frac{\mu_0 i}{2R} \odot + \frac{\mu_0 i}{4\pi R}(\sin 45^\circ + \sin 90^\circ) \odot$

$= \frac{-\mu_0 i}{4\pi R}\left[1-\frac{1}{\sqrt{2}}\right] + \frac{\mu_0 i}{2R} + \frac{\mu_0 i}{4\pi R}\left[\frac{1}{\sqrt{2}}+1\right] \odot$

$= \frac{\mu_0 i}{4\pi R}\left[-1+\frac{1}{\sqrt{2}}+2\pi+\frac{1}{\sqrt{2}}+1\right]\odot \qquad = \frac{\mu_0 i}{4\pi R}\left[\sqrt{2}+2\pi\right]\odot \qquad = \frac{\mu_0 i}{2\pi R}\left[\frac{1}{\sqrt{2}}+\pi\right]\odot$

Q7

Consider an infinitely long current carrying cylindrical straight wire having radius 'a'. Then the ratio of magnetic field at distance $\frac{a}{3}$ and 2a from axis of wire is.

एक अनन्त लम्बे धारावाही बेलनाकार सीधे तार कि त्रिज्या 'a' है। तब अक्ष से $\frac{a}{3}$ तथा 2a दूरी पर चुम्बकीय क्षेत्र का अनुपात होगा–

(1) $\frac{3}{5}$ (2) $\frac{2}{3}$ (3) $\frac{1}{2}$ (4) $\frac{4}{3}$

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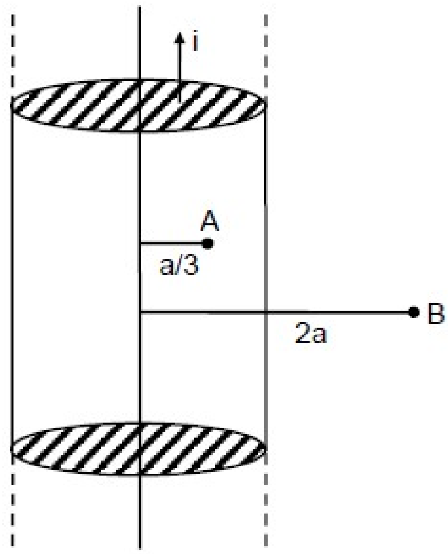
(2)



$$B_A = \frac{\mu_0 i r}{2\pi a^2} = \frac{\mu_0 i \frac{a}{3}}{2\pi a^2} = \frac{\mu_0 i}{\pi a^2}\,\frac{a}{6} = \frac{\mu_0 i}{6\pi a}$$

$$B_B = \frac{\mu_0 i}{2\pi(2a)}$$

$$\frac{B_A}{B_B} = \frac{4}{6} = \frac{2}{3}$$

Q8

There is a long solenoid of radius 'R' having 'n' turns per unit length with current i flowing in it. A particle having charge 'q' and mass 'm' is projected with speed 'v' in the perpendicular direction of axis from a Point on its axis Find maximum value of 'v' so that it will not collide with the solenoid.

एक लम्बी परिनालिका जिसकी त्रिज्या 'R' तथा इकाई लम्बाई में घेरो की संख्या 'n' है, मे धारा i प्रवाहित हो रही है। 'q' आवेश व 'm' द्रवयमान का एक कण 'v' चाल से अक्ष से अक्ष के लम्बवत् प्रेक्षेपित किया जाता है। 'v' का अधिकतम मान ज्ञात करो ताकि यह परिनालिका से नहीं टकराये –

(1) $\frac{Rq\mu_0 in}{2m}$ (2) $\frac{2Rq\mu_0 in}{m}$ (3) $\frac{Rq\mu_0 in}{3m}$ (4) $\frac{Rq\mu_0 in}{4m}$

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(1)

$$R_{max} = \frac{R}{2} = \frac{mv_{max}}{q\mu_0 in}$$

$$V_{max} = \frac{Rq\mu_0 in}{2m}$$

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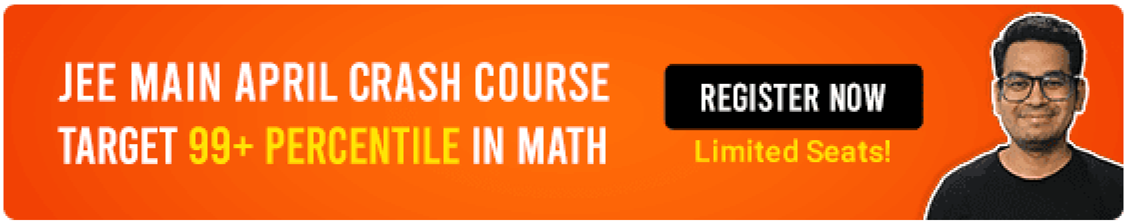

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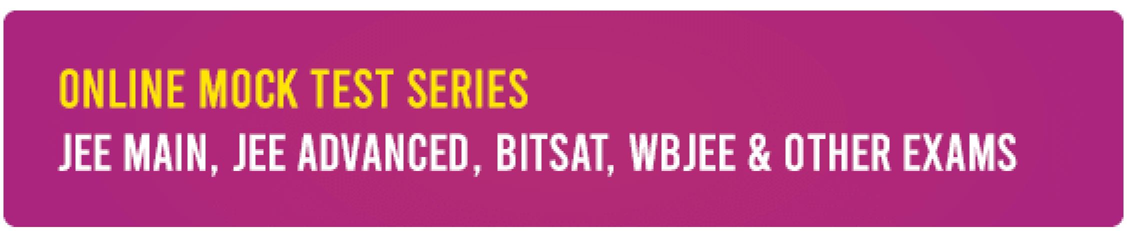

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